Embedded Systems Design Patterns - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Embedded Systems Design Patterns - NUIST (A35615)

Short Title:	Embed Systems Design Patterns (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Discipline Specific Technologies		
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Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module will cover the design and programming of embedded systems. The student will be introduced to the salient components of an embedded system and then more advanced design patterns such as Master/Slave, Distributed and RTOS applications.

Programmes

`EMBS-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / M
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	1 / 2 / M
`EMBS-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / M

Indicative Content

- Architecture of Embedded Systems
- System Expansion
- Interrupt and exception handlers
- Test, debug and verify embedded applications
- Real Time Operating System
- Web Servers Ethernet and Wireless
- Subroutines & Parameter Passing
- Development Tools
- System Integration
- Serial & Network Bootstrapping

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Design embedded MPU systems with aid of hardware and software development tools.
2. Develop Distributed Multiprocessor and Multitasking systems.
3. Set up a master slave protocol design pattern.
4. Produce Embedded applications using Real Time Operating System Design Pattern.
5. Test, Debug and Verify embedded applications.
6. Manage and maintain embedded software/hardware applications.

Learning and Teaching Methods

- Lectures
- Project Based Learning
- Case Studies

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Case Studies	50%	1,4,5,6
Case Studies	50%	1,2,3,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Independent Learning	105	

Essential Material

- "mbed." https://developer.mbed.org/
- CCS, CCS. _CCS C Compiler Manual._. www: CCS, July 2016.

Requested Resources

- ENGINEERING LAB: Electronics